

On wave propagation in altermagnets

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Cervantes

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Supervisors

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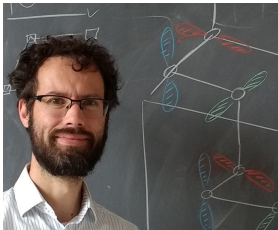
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Simulation Results

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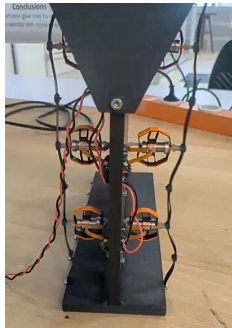
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Click the image below to play the simulation video:



Motivation

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What is the project about?

Altermagnetic symmetry/Lieb lattice +
non-hermiticity/non-reciprocity, in classical analogues

Underlying idea

Guide wave signals in solids

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What is the project about?

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Underlying idea

Guide wave signals in solids

One can use different excitations: electrons, photons, phonons, etc.

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What is the project about?

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Underlying idea

Guide wave signals in solids

One can use different excitations: electrons, photons, phonons, etc.

- 1 Anisotropic/directional
- 2 Robust $\longrightarrow$ topologically protected
- 3 Spin-momentum locking

Examples

Example 1: quantum spin Hall effect

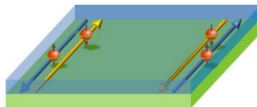


Figure: Irene Aguilera Topological materials course

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Examples

Example 1: quantum spin Hall effect

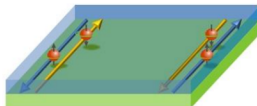


Figure: Irene Aguilera Topological materials course

Example 2: 2D array of pendula

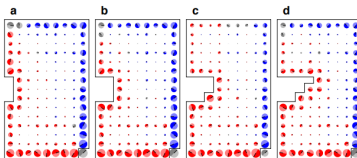


Figure: [?]

What are altermagnets? A different phase of matter I

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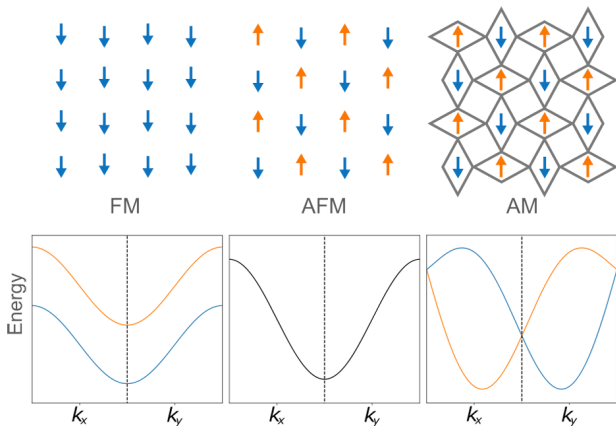


Figure: [?]

What are altermagnets? A different phase of matter II

- 1 Symmetry: Rotation+TR denoted by $C_4\mathcal{T}$
- 2 Zero magnetization
- 3 Spin-split bands
- 4 $C_4\mathcal{T} \implies N_{\uparrow} = N_{\downarrow}$

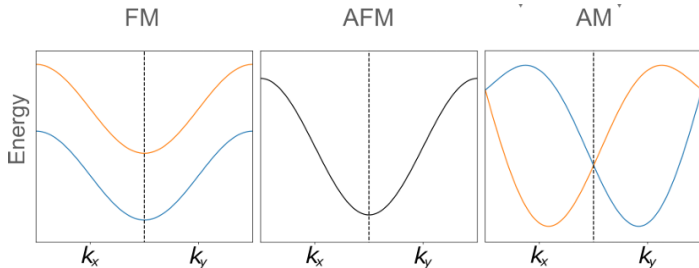


Figure: [?]

Lieb lattice as a model altermagnet

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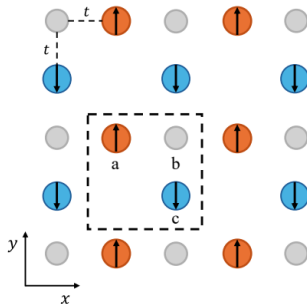
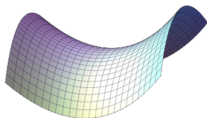


Figure: [?]

It has hyperbolic bands:



Properties of altermagnets

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- 1 Anisotropic: control wave propagation direction
- 2 Spin-momentum locking: different spins propagate in different directions
- 3 Piezomagnetism: magnetic moment is induced by strain

Useful in spintronics, spin-splitters, ultrafast photomagnetism, thermoelectrics ...

Generalization of alter symmetry

Generalization $C_4\mathcal{T}$ symmetry

spin $\longrightarrow$ 2-state variable

Some examples are:

- Photon polarization $\longrightarrow$ photonic altermagnet
- Electric dipole $\longrightarrow$ alterelectric
- Pendula chirality $\longrightarrow$ mechanical altermagnet (metamaterial)

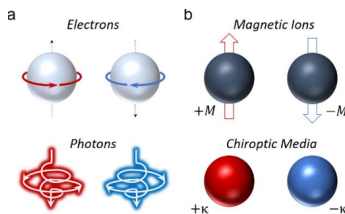


Figure: [?]

Non-hermiticity I

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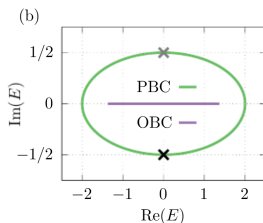
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- Motivation: gain/loss for pendula chirality, non-reciprocal hopping
- In hermitian systems: PBC bulk $\rightarrow$ OBC edge states
- In non-hermitian systems: extreme sensitivity to boundaries
- Spectrum becomes complex



Non-hermiticity II

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Figure: [?]

- **New** non-hermitian skin effect: eigenstates localize at boundary
- Example: Hatano-Nelson model

$$\hat{H}_{\text{HN}} = \sum_j \left[t_R c_{j+1}^\dagger c_j + t_L c_j^\dagger c_{j+1} \right]$$

Non-reciprocity I

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- Meaning: asymmetric interaction
- Example: diode
- Motivation: breaks inversion symmetry $\rightarrow$ directional waves
- $$\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} k & -k^o \\ +k^o & k \end{pmatrix} \begin{pmatrix} \Delta_1 \\ \Delta_2 \end{pmatrix}$$

Non-reciprocity II

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Figure: Non-reciprocal actuators

Non-reciprocity III

■ Non-reciprocal units

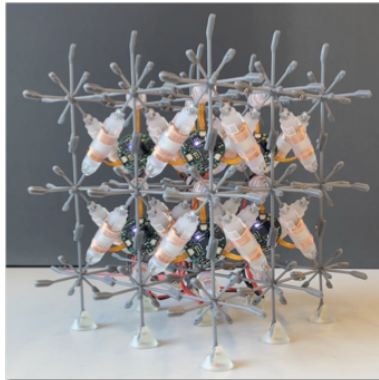


Figure: 3d cube

In practice

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What is the project about?

Altermagnetic symmetry/Lieb lattice +
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In practice

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What is the project about?

Altermagnetic symmetry/Lieb lattice +
non-hermiticity/non-reciprocity, in classical analogues

The procedure has been:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Map to quantum system (using multiple scale expansion)
- 4 Analyze spectrum and eigenstates

In practice

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What is the project about?

Altermagnetic symmetry/Lieb lattice +
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The procedure has been:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Map to quantum system (using multiple scale expansion)
- 4 Analyze spectrum and eigenstates

The idea is to apply them to increasingly more complex models.

Experimentally

Build a Lieb lattice of non-reciprocal pendula and compare with theoretical predictions

Multiple scale expansion: the map to quantum system I

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Technique useful for solving nonlinear ODEs with different timescales

Simple example: weakly damped harmonic oscillator

$$\ddot{x} + x + 2\varepsilon\dot{x} = 0$$

How do we approximate solution? With a power series in ε :

$$x = x_0(t) + \varepsilon x_1(t) + \mathcal{O}(\varepsilon^2)$$

$$\text{Substituting } [\ddot{x}_0 + x_0] + \varepsilon[\ddot{x}_1 + x_1 + 2\dot{x}_0] + \mathcal{O}(\varepsilon^2) = 0$$

$$\text{Solution: } x(t, \varepsilon) = \sin t - \varepsilon t \sin t + \mathcal{O}(\varepsilon^2)$$

Multiple scale expansion: the map to quantum system II

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Can we find an approximate solution valid for $t > \mathcal{O}(1/\varepsilon)$?

Define 2 timescales: $\tau = t$ and $T = \frac{t}{\varepsilon}$

Rewrite solution: $x(t, \varepsilon) = x_0(\tau, T) + \varepsilon x_1(\tau, T) + \mathcal{O}(\varepsilon^2)$

Time derivative: $\dot{x} = \partial_\tau x_0 + \varepsilon(\partial_T x_0 + \partial_\tau x_1) + \mathcal{O}(\varepsilon^2)$

$\mathcal{O}(\varepsilon^0)$: $\partial_{\tau\tau} x_0 + x_0 = 0 \longrightarrow x_0 = A(T) \sin \tau + B(T) \cos \tau$

$\mathcal{O}(\varepsilon)$: $\partial_{\tau\tau} x_1 + x_1 = -2(\partial_T x_0 + \partial_\tau x_0)$

We require no secular terms $\longrightarrow A' + A = 0 \quad B' + B = 0$

Approximate solution: $x = e^{-\varepsilon t} \sin t + \mathcal{O}()$

Note: this technique is much more general.

Classical to quantum

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Assumption: treat non-harmonic terms weakly $\longrightarrow \varepsilon$

Oscillating solutions with (slow) time-dependent coefficients

The coefficients evolve as $\vec{A}' = M\vec{A}$

Map to a Schrödinger equation with Hamiltonian $H = iM$

Use CMT knowledge to analyse system: spectrum and eigenstates

Non-reciprocal pendulum

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$$\ddot{x} + kx + \epsilon\gamma\dot{x} + \epsilon k^a y = 0 \quad (1)$$

$$\ddot{y} + ky + \epsilon\gamma\dot{y} - \epsilon k^a x = 0 \quad (2)$$

The energies are

$$\text{clockwise - decay } E_1 = -\frac{i}{2} \left(\gamma + \frac{k^a}{\sqrt{k}} \right) \quad (3)$$

$$\text{anticlockwise - grow } E_2 = \frac{i}{2} \left(-\gamma + \frac{k^a}{\sqrt{k}} \right) \quad (4)$$

E_1 has 2 eigenvectors rotating clockwise with different initial phase- these modes decay. E_2 has 2 eigenvectors that rotate anticlockwise with different initial phase- these modes grow.

2 coupled non-reciprocal pendula

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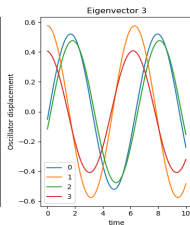
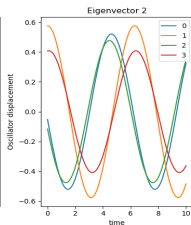
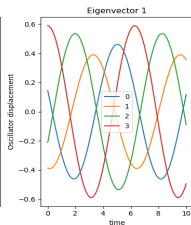
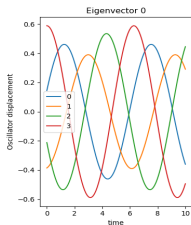
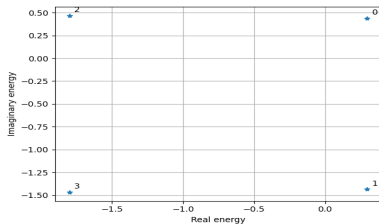
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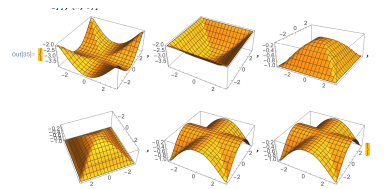
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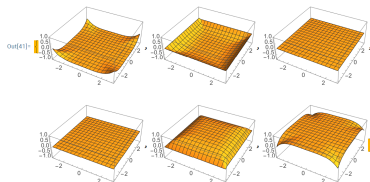


Lieb lattice of non-reciprocal pendula I

It seems like there are modes with zero energy in certain directions Real spectrum:



Imaginary spectrum:



Hatano-Nelson ladder PBC

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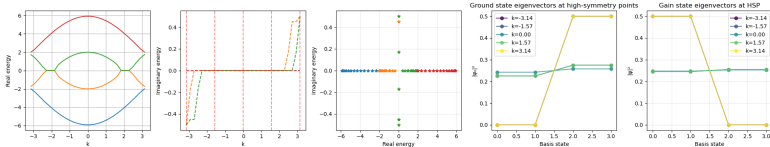


Figure: Hatano-Nelson ladder PBC with imaginary onsite potential.

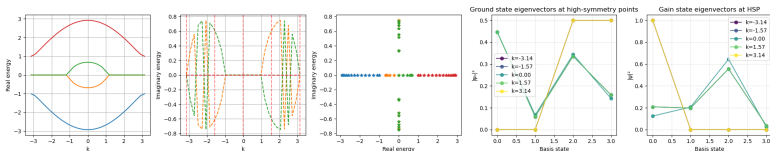


Figure: Hatano-Nelson ladder PBC with non-reciprocal hopping.

Hatano-Nelson ladder OBC I

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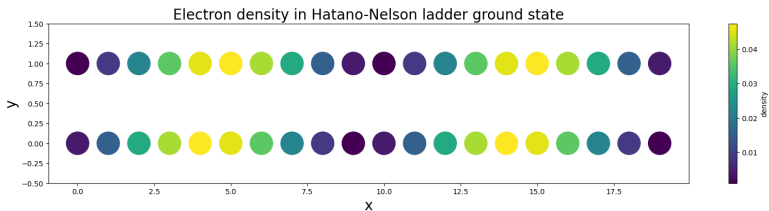


Figure: Hatano-Nelson ladder OBC ordinary: $U=0$ and reciprocal hopping.

Hatano-Nelson ladder OBC II

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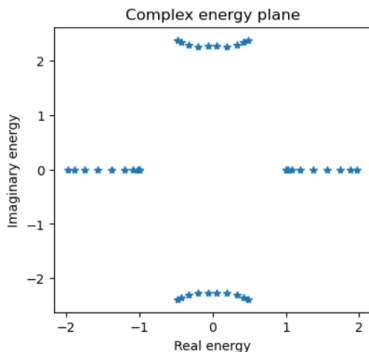


Figure: Spectrum of Hatano-Nelson ladder OBC with imaginary onsite potential.

Hatano-Nelson ladder OBC III

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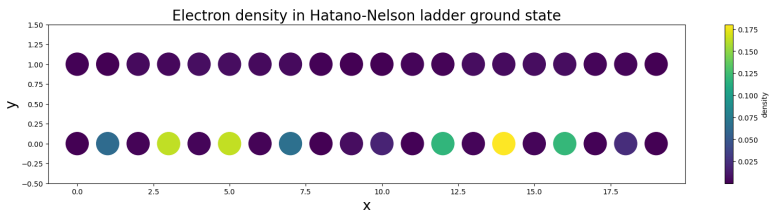


Figure: Hatano-Nelson ladder OBC with imaginary onsite potential.

Hatano-Nelson ladder OBC IV

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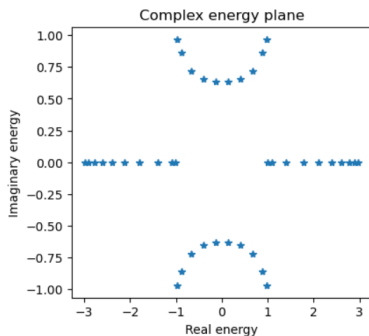


Figure: Spectrum of Hatano-Nelson ladder OBC with non-reciprocal hopping.

Hatano-Nelson ladder OBC V

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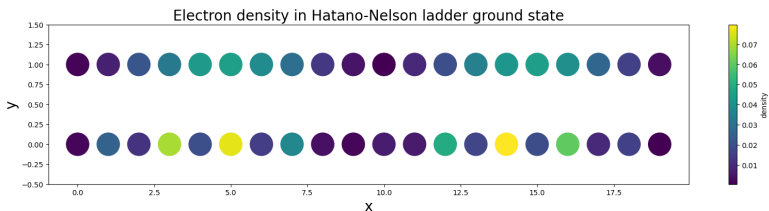


Figure: Hatano-Nelson ladder OBC with non-reciprocal hopping.

Ongoing: bilayer Lieb lattice I

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Stack Lieb lattices

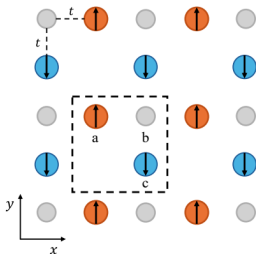


Figure: [?]

Ongoing: bilayer Lieb lattice II

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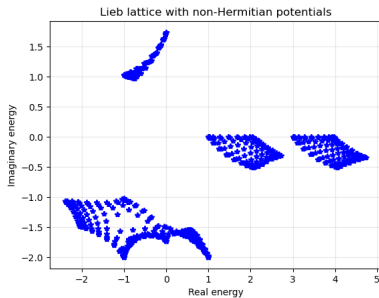


Figure: Spectrum of bilayer Lieb lattice with imaginary onsite potential.

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Goal

Study altermagnetism and non-hermiticity in classical analogues

- Altermagnets: $C_4\mathcal{T}$ symmetry can lead to anisotropic and spin-momentum locking propagation
- Non-hermitian skin effect: waves localized at boundaries
- Classical analogues enable experimental realization

Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.

Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.

There are different ways to achieve activity:

Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping

Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping
- Time delay

Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping
- Time delay
- **Non-reciprocity**

2 interacting non-reciprocal pendula

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Out[58]//MatrixForm=

$$\begin{pmatrix} -\gamma & -\beta & 0 & \sigma & 0 & \beta & 0 & 0 \\ \beta & -\gamma & \sigma & 0 & -\beta & 0 & 0 & 0 \\ 0 & -\sigma & -\gamma & -r & 0 & 0 & 0 & r \\ -\sigma & 0 & r & -\gamma & 0 & 0 & -r & 0 \\ 0 & -\beta & 0 & 0 & -\gamma & \beta & 0 & \sigma \\ \beta & 0 & 0 & 0 & -\beta & -\gamma & \sigma & 0 \\ 0 & 0 & 0 & -r & 0 & -\sigma & -\gamma & r \\ 0 & 0 & r & 0 & -\sigma & 0 & -r & -\gamma \end{pmatrix}$$

In[59]:= **Eigenvalues**[M]

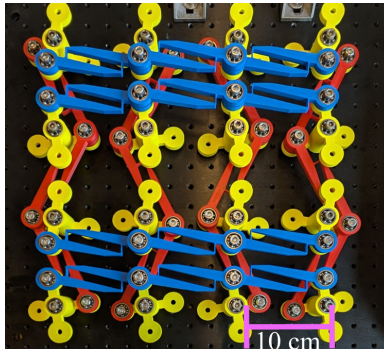
Out[59]= $\{-\gamma - i\sigma, -\gamma - i\sigma, -\gamma - i\sigma, -\gamma - i\sigma, -\gamma + i\sigma, -\gamma + i\sigma, -\gamma + i\sigma, -\gamma + i\sigma\}$

In[60]:= **Eigenvalues**[T2]

Out[60]= $\{-i\beta, i\beta, -ir, ir\}$

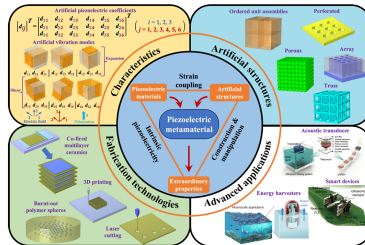
Mechanical altermagnet

- Focus on anisotropic wave propagation feature of altermagnets
- Exp: build on Amber's work: stack 2 toy layers
- Th: construct model and explain observations
- Study propagation of waves
- Main source: Amber's thesis



Piezoelectric

- Exp: build altermaterial using capacitors with piezoelectric response
- Th: construct microscopic quantum model and compare
- Auxetic piezo
- Best case scenario: show piezoelectric properties of alterelectrics
- Review by Gao, Sharma



Non-hermiticity

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- Combine altermagnetic symmetry with non-hermiticity
- Exp: stack Yuan's cubes with odd elasticity
- Study waves propagating
- Gain-momentum locking
- Th: write analogous quantum hamiltonian and analyse it



Other

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- Metamaterials with higher order altermagnet symmetries
- Consider other rotations besides 90°
- For theory in general: band structure, ground state, excitations, topological properties, defects

Research plan

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- Calculate band structure of mechanical Lieb lattice
- Simulate Lieb lattice of non-reciprocal units
- Build Lieb lattice of non-reciprocal units

Possible questions

- Are there skin effects?

Do altermagnets always have hyperbolic dispersion?

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Do altermagnets always have hyperbolic dispersion?

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1 Break TRS

Do altermagnets always have hyperbolic dispersion?

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- 1 Break TRS
- 2 C4T symmetry

Do altermagnets always have hyperbolic dispersion?

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- 1 Break TRS
- 2 C4T symmetry
- 3 Inversion symmetry

Do altermagnets always have hyperbolic dispersion?

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- 1 Break TRS
- 2 C4T symmetry
- 3 Inversion symmetry
- 4 Zero magnetisation

Hatano-Nelson ladder

Parameters: $U = 1$, $r = 2$, $t = 1$, $a = 1$

